

18 > Monitor, maintain & future-proof

IV spots open pairs, near-end crosstalk and 100 m over-length inside 30 seconds.

Field drill

1. Plug tester into suspected port
2. Run “Wiremap+TDR”
3. If length >95 m or NEXT > 4 dB, re-terminate or re-route

Log faults to LibreNMS via SNMP trap so future audits know the fix date.

Patch, Backup, Repeat: Automation scripts & gold-image recovery

The 48-hour patch pledge

CERT-In's April 2023 advisory cites “unpatched firmware” in 52 % of SME breaches. Automate or drown.

Ansible one-liner for Linux Patch Tuesday

Red Hat shows how a 12-line playbook patches hundreds of Ubuntu and RHEL boxes, reboots only if kernel updated, and Slack-notifies when done.

```
---
- hosts: linux
  become: yes
  tasks:
    - name: Update & reboot when needed
      yum:
        name: '*'
        state: latest
        register: result
    - reboot:
      when: result.changed
```

Schedule via `ansible-pull` cron on the control box; logs ship to Zabbix with `zabbix_sender`.

Windows autopilot reset as Gold-Image

Forget PXE rooms. Windows Autopilot Reset reverts a laptop to a business-ready state in under 15 minutes, preserving AzureAD join and MDM enrollment – perfect for RMA swaps or ransomware nukes.

Workflow:

1. IT clicks Fresh Start in Intune.
2. Device downloads 4 GB image, reboots twice, lands on OOB screen.
3. New user signs in; Intune re-installs apps and VPN profile.

Automate switch-firmware upgrades with Ansible

Hand-uploading `.bin` images to two dozen switches is how Friday nights disappear. A community playbook on Medium

shows the pattern: push image, set boot var, reload on your schedule.

```
---
- name: Upgrade IOS XE on access layer
  hosts: access
  gather_facts: no
  tasks:
    - name: Copy image
      ansible.netcommon.scp:
        src: cat9300-universalk9.17.09.04.SPA.bin
        dest: flash:
    - ios_config:
      lines:
        - boot system switch all flash:cat9300-universalk9.17.09.04.SPA.bin
    - ios_command:
      commands: reload in 1
```

Run it with `ansible-playbook -i inventory upgrade.yml --limit access` and go home while cron handles reboots.

Config backups that version-control themselves

LibreNMS integrates Oxidized, an open-source backup engine that yanks running-configs over SSH/SNMP, drops them in a Git repo, and auto-diffs every change. One click in Global Settings ► External ► Oxidized keeps the node list synced; a rogue CLI tweak after 9 p.m. surfaces in Git history by 9:05. Admins swapping Oxidized swear by it over RANCID because commits are immediate and human-readable.

Drill: Pull a bad VLAN command, run `oxidized-script -git checkout <hash>` to restore, copy-paste back into the switch, and commit to a good state.

Table-top disaster drill: 90 minutes, zero cost

A convincing disaster rehearsal beats dusty PDFs. AlertMedia's six-step template lays out objectives, inject cards, and hot-wash debrief. Pair it with Nexight Group's nine-step design checklist to pick roles and ground rules.

1. **Scenario:** ISP fibre cut + ransomware on file server.
2. **Inject 1:** Edge router drops BGP; NMS screens red-wall after 2 min.
3. **Inject 2:** CFO's laptop encryption notice.
4. **Goal:** Restore Internet on 4G failover, wipe server, push Autopilot gold image, full ops inside 60 min.
Videotape, chronometer, then refine SOP gaps before the next real outage.

CERT-In's 48-hour patch mandate

India's cyber-agency repeatedly urges SMEs to apply vendor patches within two days of release. Older guidelines in the